Instrunction set 0f 8051 divided in 5 groups
Data transfer operation - MOV, MOVC,MOVX,XCHG,XCHD
Arithmetic Operation - ADD, ADDC,SUBB, MUL,DIV,INC,DEC
Logical Operation - ANL,ORL,XRL (Anding, ORing,XORing Operation)
Rotation operation, SWAP operation
Brancing Operation - conditional and Unconditional Jump Operation, CALL and RET operation
Bit manipulation Operation - applicable only for bit.

Example1
```
org    0000h
;addition without carry
        mov a, #06h
        mov b, #09h
        add a, b
        mov r0, a
;Substraction with carry
        mov a, #08h
        subb a, #03h
        mov r1, a
;multiplication
        mov a, #03h
        mov b, #06h
        mul ab
        mov r2, a
;division
        mov a, #08h
        mov b, #03h
        div ab
        mov r3, a
        mov r4, b
;increment operation
        mov a, #03h
        inc a
        mov r5, a
;decrement operation
        mov r6, #07h
        dec r6
        lcall 0003h
        end
```

Example 2
```
org    0000h
        mov r0, #0fh
        mov r1, #0f0h
        mov r2, #66h
; And operation
        mov a, #ffh
        anl a, r0
        mov r3, a
; Or operation
        mov a, #ffh
        orl a, r1
        mov r4, a
```

```
; Xor operation
	mov a, 03h
	mov a, #ffh
	xrl a, r2
	mov r5, a
	lcall 0003h
	end
```

Example 3

```
org    0000h
; clear register A
	mov a, #0fh
	clr a
	mov r0, a
; swap nibbles of register A
	mov a, #56h
	swap a
	mov r1, a
; Complement the bit of register A
	mov a, #66h
	cpl a
	mov r2, a
; Rotate the register contents towards right
	mov a, #63h
	rr a
	xrl a, r
	mov r3, a
; Rotate the register contents towards left
	mov a, #43h
	rl a
	xrl a, r
	mov r4, a
	lcall 0003h
	end
```

Bit manipulation operations:

Example 4

```
org    0000h
	mov a, #0ffh
	clr c
;clear the carry flag
	anl c, acc.7
	mov r0, a
	setb c
;set the carry flag
	mov a, #00h
	orl c, acc.5
	mov r1, a
	mov a, #0ffh
	cpl acc, 3
	mov r2, a
	lcall 0003h
	end
```